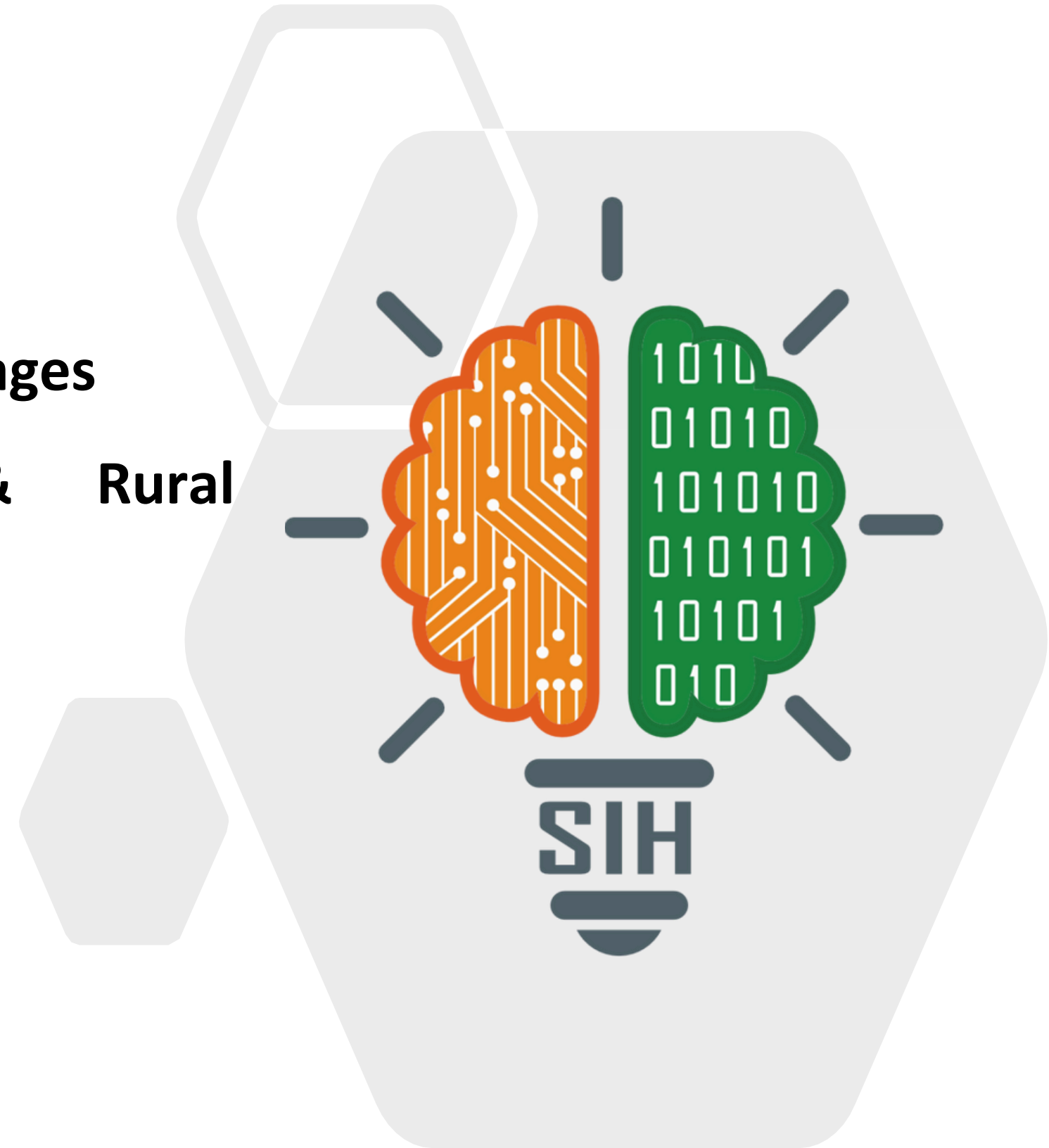
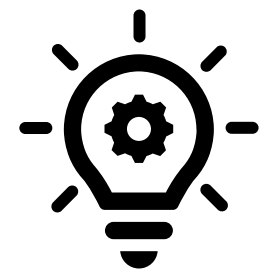


SMART INDIA HACKATHON 2024

- Problem Statement ID - SIH25126
- Problem Statement Title – Zero Food Wastages
- Theme – Agriculture, FoodTech & Rural Development
- PS Category - Software
- Team ID - 91282
- Team Name - Bug Busters x6



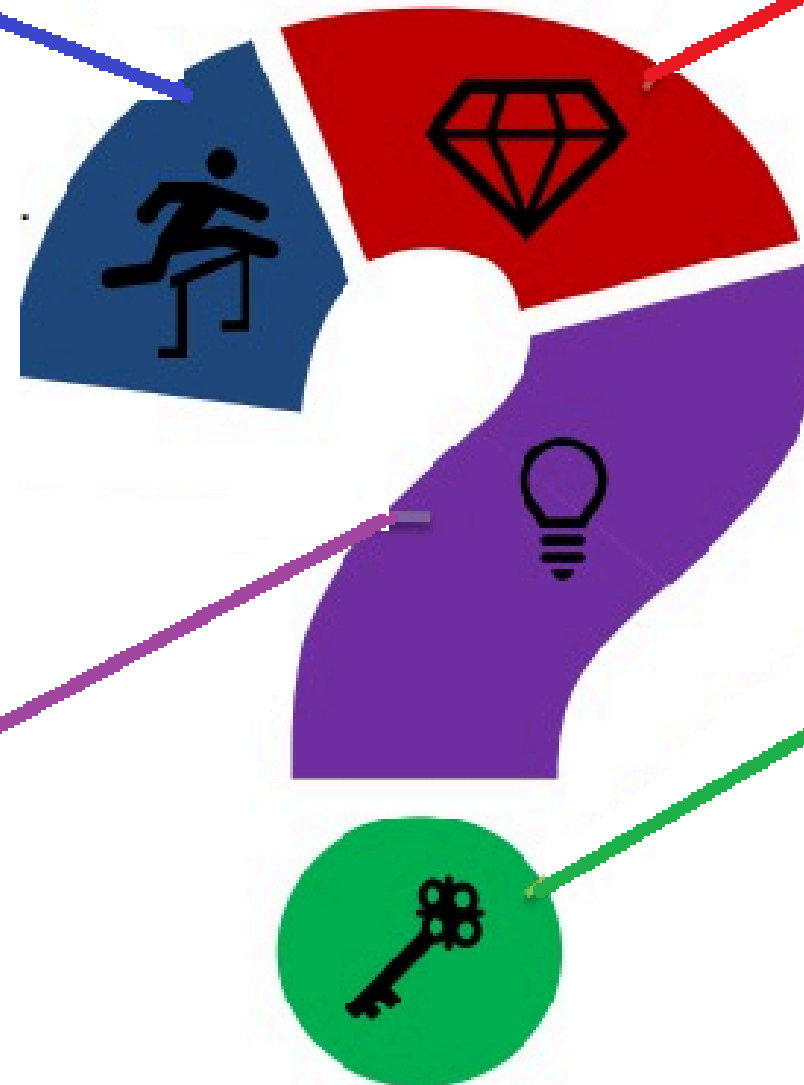


Challenges

- Real-time food availability update
- Integration of payment & location services
- Food quality & safety verification
- Collaboration with restaurants/NGOs
- Logistics for pickup & delivery
- Building user trust & awareness

Proposed Solution

- Distribute surplus food to needy people, ensuring zero hunger.
- Convert leftover food waste into electricity through biogas plants.
- Use AI-based checks to maintain food quality and safety.
- Create social and environmental impact with one smart solution.



Value proposition

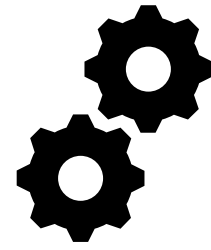
- Zero Hunger, Zero Waste
- Food to People, Waste to Power
- Social Good + Green Energy
- Safe, Smart, Sustainable
- Impact Beyond Meals

Key Features

- Food Collection & Redistribution
- Real-Time Tracking
- Donor & Volunteer Management
- Food Safety & Quality Check
- Waste to Resource

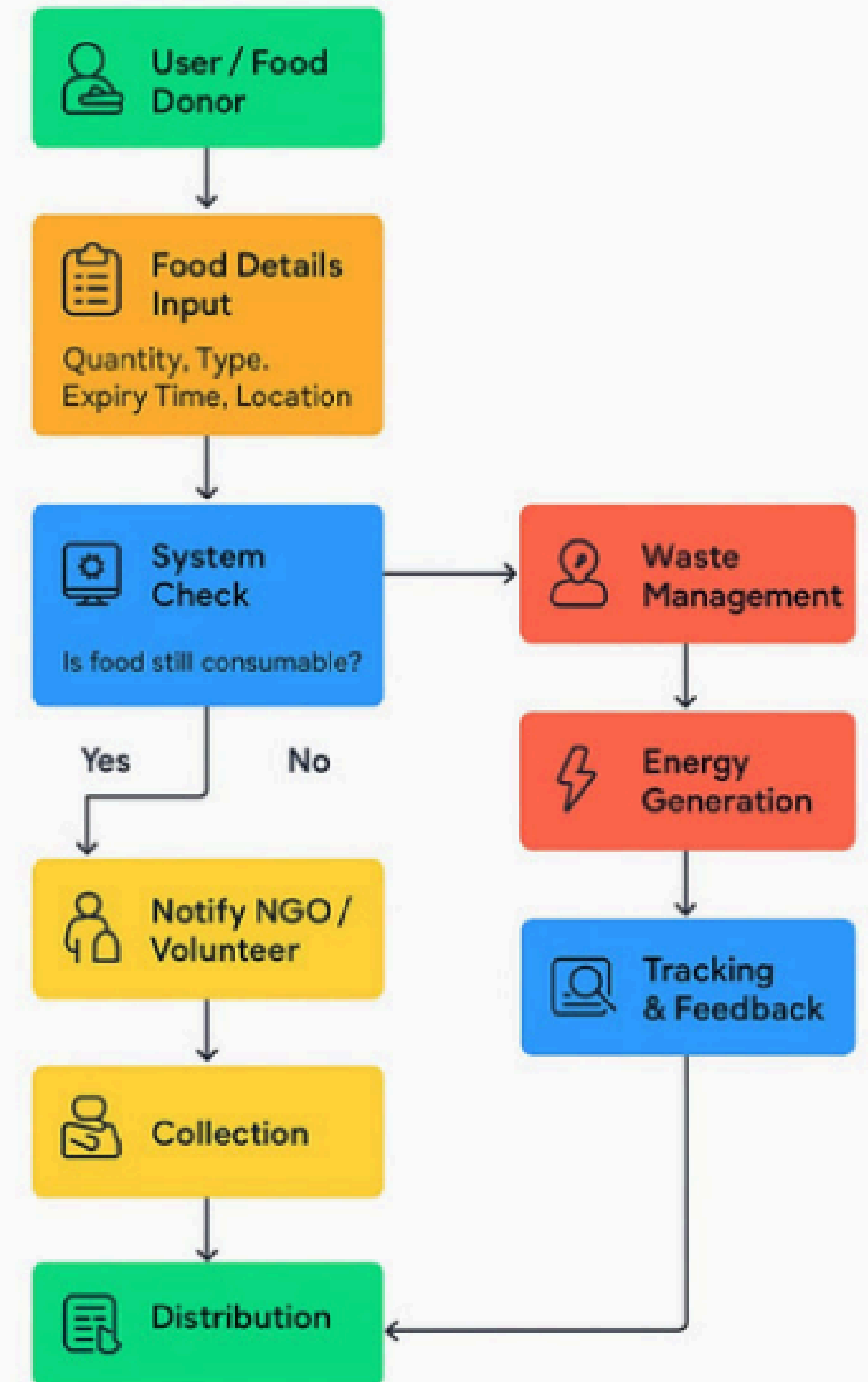
Uniqueness

- AI-Powered Demand Prediction
- Expiry & Portion Control Alerts
- Transparency with Dashboard
- Reward & Recognition Model



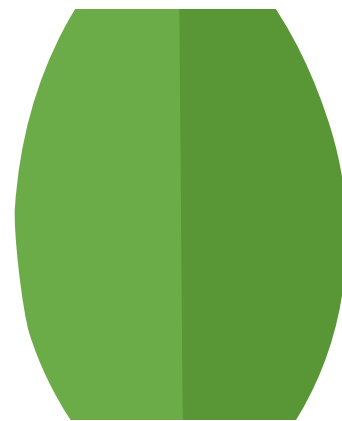
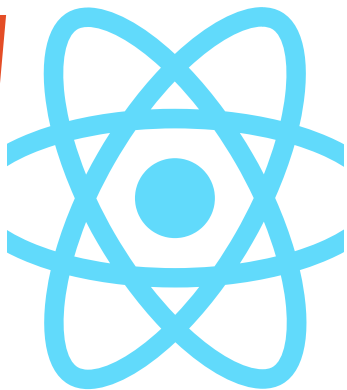
- **Frontend:** HTML, React.js, Bootstrap CSS for modern responsive UI
- **Backend:** Node.js for API handling, Python
- **Database:** MongoDB for structured cultural data
- **Hosting & Deployment:** Netlify for scalable cloud hosting

Proposed Solution



View Prototypes :

<https://fluffy-dusk-e20b2d.netlify.app/>



Analysis of the feasibility of the idea:

- Existing frameworks support mobile & web-based development.
- Scalable across cities, with future nationwide/global expansion potential.
- User-friendly interface for donors, NGOs, volunteers, and households.
- Promotes social responsibility and reduces hunger & environmental impact.

Challenge 1: Food Collection & Logistics

Difficulty in collecting surplus food on time due to lack of proper coordination between donors, volunteers, and NGOs.

Solution

Implement a real-time donor-volunteer matching system using GPS.
Partner with local delivery agencies/volunteers for faster pickups.
Introduce scheduling for large food donations from hotels & events.

Challenge 2: Food Safety & Quality

Risk of food being spoiled, unhygienic, or unsafe for consumption during collection, storage, and redistribution.

Solution

Develop guidelines for safe food handling & packaging.
Use IoT-enabled temperature & freshness sensors for storage.
Partner with FSSAI/health departments to ensure safety standards.

Challenge 3: User Adoption & Awareness

Restaurants, households, and individuals may hesitate to adopt the platform due to lack of awareness or motivation.

Solution

Launch campaigns, workshops, and digital drives to spread awareness.
Provide rewards, recognition, and CSR benefits for regular donors.
Highlight impact via dashboards showing meals saved & people fed.

01 Large Target Audience:

India generates ~67 million tons of food waste annually, while millions of poor and homeless people can directly benefit from surplus food redistribution.

02 Hunger & Malnutrition Reduction:

Supplying surplus food to the needy helps reduce hunger and malnutrition, especially among children and women.

03 Environmental Impact:

Minimizing food waste reduces landfill garbage and methane/CO₂ emissions → leading to a cleaner environment and climate change mitigation.

04 Government & NGO Support:

Aligned with initiatives like Swachh Bharat Mission and FSSAI food safety drives, with strong potential for NGO collaboration and CSR funding.

05 Market & Social Opportunity:

Digital platforms enable restaurants, hostels, and event organizers to manage waste sustainably, while reward and recognition systems motivate continuous participation.

Challenges Faced:

- According to the United Nations Environment Programme (UNEP), India wastes around 67 million tons of food every year, worth nearly ₹92,000 crores, even while millions go hungry.
- Reports highlight that 40% of the food produced in India is wasted, mainly during weddings, events, restaurants, and supply chains.
- Lack of awareness, poor logistics, and absence of proper redistribution systems worsen the food wastage problem.

REFERENCE PDFS BY CIS BENCHMARK:

- [No Food Waste - Not-For-Profit surplus food management organisation](https://globalgoals.org/goals/2-zero-hunger/)
- <https://globalgoals.org/goals/2-zero-hunger/>
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